Programming for Robotics – Final Group Project (70%)

Design, implement, and demonstrate a robotic system integrating concepts learned throughout the semester.

1. Important Dates

- **Grouping completed in LMS:** Sunday, **Week 3** (10% deduction if not completed)
- **Pitch document due:** Sunday, **Week 6** (20% deduction if not completed)
- **Final submission:** Monday **9:00am, Week 12** (10% deduction per day late)
- **Presentation/video review** during **Week 12** workshops (*no presentation → No mark)*

2. Group Structure

- Maximum **3 students per group**
- All members must:
  - Contribute technically
  - Speak in the submission video
  - Answer questions about system design and code in Week 12

3. Submission Details

- **Pitch document**: max 2 pages in MS Word, due by Week6.
- **Final Project Report:** max 4 pages in MS Word, strictly following the provided project **report template**.
- All **source code** and the project demonstration **video** must be uploaded to a **private GitHub** repository. Each group must invite the 3 teaching staff assessors as collaborators to allow access for marking purposes. The GitHub **repository link** must be clearly included in the submitted **report**.
- In the report, explicitly explain how your system satisfies all '**Technical Requirements**', the below.

4. Technical Requirements

Choose one of the options below. Your project must demonstrate the following core components:

Core Requirements (Apply to ALL Projects)

All projects must include:

- At least 2 **inputs** (sensors or perception sources)
- At least 2 **outputs** (actuators or behaviour outputs)
- **FSM** or **Behaviour Tree**
- Minimum 4 **behavioural states**
- **Multi-condition** decision logic
- **Safety** or **fail-safe** mechanism
- Structured **system architecture** diagram

Then choose **ONE** specialisation from below:

A. Embedded Intelligence (Wokwi)

- Sensor fusion

- Context-aware behaviour
- Real-time logic

B. Autonomous Robotics (Webots)

- Navigation
- Obstacle avoidance
- Perception-driven decision

C. Multi-Agent Coordination (Webots)

- At least one coordination mechanism
- Inter-agent communication
- Conflict resolution

D. Human–Robot Interaction (Webots)

- Structured interaction flow
- Perception-triggered response
- Smooth behaviour transitions

5. Video Submission (Maximum 5 Minutes)

- **Introduction (project + members):** 1 minute
- **Demonstration:** 3 minutes
- **Code & Architecture Explanation:** 1 minute

Total maximum duration: **5 minutes**

Video Requirements

- Every member must speak
- Show system working clearly
- Show code editor briefly
- Show architecture diagram
- Demonstrate intelligent/adaptive behaviour

6. Code Submission

Code must be **modular** and **structured**, clearly **commented**, logically **organised** and meaningfully **named.**
Students must be able to explain, control flow, state transitions, safety logic, and aI or coordination logic

6. Use of ChatGPT

You may use AI Tools however:

- You must fully understand all submitted code.
- You must be able to explain design decisions.

If a student cannot answer reviewers' questions → **marks will be reduced individually.**

8. Marking Criteria

8.a. Core Criteria (Applicable to ALL Projects)

The following criteria apply to every project, regardless of system type:

- **Technical Requirements & System Integration** – 30%
  The project correctly implements and integrates all required technical components of the chosen option.
- **System Architecture** – 15%
  Clarity and coherence of the overall system structure and separation of sensing, processing and actuation.
- **Behaviour & Control Logic** – 15%
  Quality and correctness of structured behaviour modelling (e.g., FSM, Behaviour Tree, or control logic).
- **Safety, Robustness & Reliability** – 10%
  Inclusion of appropriate safety mechanisms and stable operation under different conditions.
- **Code Quality & Engineering Practice** – 10%
  Modularity, readability, organisation, and overall quality of the submitted code.
- **Video Demonstration** – 5%
  Clarity and effectiveness of the system demonstration within the required time limit.
- **Individual Technical Understanding** – 5%
  Each member's ability to clearly explain system design and technical decisions.

Subtotal: 90%

8.b. Advanced / Contextual Component (10%)

Students must demonstrate **at least one advanced element** appropriate to their chosen project type. This component contributes **10%** to the final mark.

- AI or Adaptive Behaviour
- Autonomous Navigation
- Multi-Agent Coordination
- Perception (Vision-Based)
- Human–Robot Interaction (HRI) Design

Final Total: 100%

9. What Assessors Focus On

Assessors will primarily evaluate:

- Correct implementation of the chosen technical requirements
- Clear system architecture and structured Sense–Think–Act design
- Well-modelled behaviour logic (e.g., FSM / Behaviour Tree / control structure)
- Meaningful intelligent or adaptive behaviour
- Appropriate safety and robustness mechanisms
- Quality of code structure and engineering practice
- Each member's technical understanding and ability to justify design decisions

Assessors will **not** focus on:

- Graphic quality or visual polish
- Complex animations without technical depth
- Lengthy presentation slides
- Cosmetic features that do not improve system intelligence

Build a system that demonstrates intelligence, not just motion.